

CityCare Multispecialty Hospital

Synthetic Clinical Case Report For RAG Testing

****Document type:**** Fictional medical case report
****Purpose:**** Retrieval-Augmented Generation testing only
****Generated for:**** MediVault AI sample document evaluation
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Patient Registration Details

Patient name: Ananya Rao
Age/Sex: 46 years / Female
Hospital ID: CCH-2026-05-1148
Visit type: Outpatient consultation with observation
Date of visit: 18 May 2026
Consulting department: Internal Medicine
Consultant: Dr. Meera Iyer, MD Internal Medicine
Reason for visit: Recurrent headaches, elevated blood pressure readings, fatigue, and intermittent dizziness.

Chief Complaints

1. Recurrent headache for 3 months, worse during stressful workdays.
2. Intermittent dizziness for 2 weeks.
3. Fatigue and poor sleep for 1 month.
4. Home blood pressure readings repeatedly above 150/90 mmHg.
5. Occasional blurred vision during headache episodes.

The patient denied chest pain, severe shortness of breath, fainting, seizure, weakness on one side of the body, slurred speech, or sudden worst-ever headache at the time of consultation.

History Of Present Illness

The patient reported dull pressure-like headaches occurring 3 to 4 times per week. Headaches were usually located in the forehead and back of the head. Pain intensity was described as 5 to 6 out of 10. The headache was sometimes associated with neck stiffness after long hours of computer work. There was no vomiting, no photophobia, no seizure, and no loss of consciousness.

She measured her blood pressure at home using an automated device. Readings over the previous 2 weeks ranged from 148/92 mmHg to 166/98 mmHg. She had not previously been diagnosed with hypertension and was not taking blood pressure medication.

The patient reported increased work stress, reduced physical activity, irregular meals, and sleeping approximately 5 hours per night. She consumed tea 4 to 5 times per day and added extra salt to snacks. She did not smoke and consumed alcohol rarely.

Past Medical History

- No known diabetes mellitus.
- No known chronic kidney disease.
- No prior stroke or heart attack.
- History of migraine-like headaches during college, currently less typical of migraine.
- Mild seasonal allergic rhinitis.
- No known thyroid disease.
- No previous major surgery.

Family History

- Father diagnosed with hypertension at 52 years of age.
- Mother has type 2 diabetes mellitus.
- Elder brother has dyslipidemia.
- No known family history of early sudden cardiac death.

Medication History

The patient was not taking any regular prescription medication. She occasionally used over-the-counter paracetamol 500 mg for headache, usually 1 tablet once or twice per week. She denied frequent use of non-steroidal anti-inflammatory drugs. She reported no known drug allergies.

Lifestyle And Diet History

The patient works as a project manager and spends 8 to 10 hours per day seated. Physical activity was limited to short walks twice per week. Diet included frequent packaged snacks, pickles, salted nuts, and restaurant meals. Daily water intake was approximately 1.5 liters. Sleep was irregular due to late work calls.

Lifestyle risk factors identified during consultation included high sodium intake, physical inactivity, stress, inadequate sleep, and family history of hypertension.

Physical Examination

General appearance: Alert, oriented, comfortable at rest.

Height: 158 cm

Weight: 72 kg

Body mass index: 28.8 kg/m²

Pulse: 84 beats/minute, regular

Respiratory rate: 16 breaths/minute

Temperature: 98.4 F

Oxygen saturation: 98% on room air

Blood pressure reading 1: 162/96 mmHg, right arm, sitting

Blood pressure reading 2: 158/94 mmHg, left arm, sitting

Repeat after 10 minutes rest: 154/92 mmHg, right arm, sitting

Cardiovascular examination revealed normal heart sounds with no audible murmur. Respiratory examination was clear bilaterally. Abdomen was soft and non-tender. Neurological examination showed normal speech, normal power in all limbs, normal gait, and no focal neurological deficit. Fundus examination was advised but not performed during the

visit.

Initial Clinical Impression

The clinical impression was newly detected stage 1 to stage 2 hypertension based on repeated clinic and home blood pressure readings. Headache was considered likely related to elevated blood pressure, stress, poor sleep, and musculoskeletal strain. There were no emergency neurological features during examination.

The patient was advised that hypertension can be asymptomatic and still increase long-term risk of stroke, heart disease, kidney disease, and vision complications if untreated.

Differential Diagnosis

1. Primary essential hypertension.
2. Stress-related tension-type headache.
3. Migraine without aura, less likely based on current pattern.
4. Thyroid dysfunction contributing to fatigue.
5. Anemia or vitamin deficiency contributing to fatigue.
6. Secondary hypertension, to be considered if blood pressure remains uncontrolled or if abnormal investigations are found.

Investigations Ordered

The following investigations were advised:

Test	Purpose
Complete blood count	Assess anemia or infection
Fasting blood glucose and HbA1c	Screen for diabetes
Lipid profile	Assess cardiovascular risk
Serum creatinine and eGFR	Assess kidney function
Serum electrolytes	Check sodium and potassium
Thyroid-stimulating hormone	Evaluate thyroid-related fatigue
Urine routine and urine albumin-creatinine ratio	Screen kidney involvement
Electrocardiogram	Baseline cardiac assessment
Fundus examination	Assess hypertensive eye changes

Investigation Results

Complete blood count: Hemoglobin 12.4 g/dL, total leukocyte count 7,800/mm³, platelet count 286,000/mm³.

Fasting blood glucose: 108 mg/dL.

HbA1c: 5.9%, suggestive of prediabetes range.
Total cholesterol: 214 mg/dL.
LDL cholesterol: 138 mg/dL.
HDL cholesterol: 43 mg/dL.
Triglycerides: 168 mg/dL.
Serum creatinine: 0.82 mg/dL.
eGFR: 92 mL/min/1.73 m².
Sodium: 139 mmol/L.
Potassium: 4.2 mmol/L.
TSH: 2.7 mIU/L.
Urine routine: No protein, no glucose, no ketones.
Urine albumin-creatinine ratio: Within normal range.
ECG: Normal sinus rhythm, no acute ischemic changes.

Final Assessment

1. Newly diagnosed hypertension with repeated elevated readings.
2. Overweight status with BMI 28.8 kg/m².
3. Prediabetes range HbA1c.
4. Borderline high LDL cholesterol.
5. Recurrent non-emergency headache likely multifactorial.
6. Lifestyle risk factors: high sodium diet, sedentary routine, stress, and inadequate sleep.

Treatment Plan

The patient was advised lifestyle modification and medication initiation after discussion of risks and benefits.

Medication plan:

- Amlodipine 5 mg once daily in the morning.
- Paracetamol 500 mg as needed for headache, maximum as per clinician guidance.
- Avoid frequent unsupervised painkiller use.
- No antibiotic or steroid was indicated.

Lifestyle plan:

1. Reduce salt intake and avoid packaged salty snacks.
2. Follow a DASH-style diet rich in vegetables, fruits, legumes, whole grains, and low-fat dairy.
3. Walk briskly for at least 30 minutes on most days of the week after medical clearance.
4. Reduce tea/cafeine intake gradually.
5. Maintain a regular sleep schedule of 7 to 8 hours.
6. Practice stress-reduction methods such as breathing exercises or scheduled breaks.
7. Target gradual weight loss of 5% to 7% over 6 months.

Home Blood Pressure Monitoring Instructions

The patient was instructed to measure blood pressure at home using a validated upper-arm device.

Instructions:

1. Sit quietly for 5 minutes before measurement.
2. Keep feet flat on the floor and back supported.
3. Place cuff on bare upper arm at heart level.
4. Avoid caffeine, exercise, and smoking for 30 minutes before measurement.
5. Take two readings one minute apart in the morning and evening for 7 days.
6. Record date, time, systolic pressure, diastolic pressure, pulse, and symptoms.

The patient was asked to bring the blood pressure log to follow-up.

Red Flag Advice

The patient was advised to seek urgent medical care if any of the following occur:

- Chest pain or pressure.
- Severe shortness of breath.
- Sudden weakness or numbness on one side of the body.
- Difficulty speaking or facial drooping.
- Fainting, seizure, or confusion.
- Sudden severe worst-ever headache.
- Persistent vomiting with severe headache.
- Very high blood pressure reading with symptoms.
- Vision loss or new severe visual disturbance.

Follow-Up Plan

Follow-up was advised after 2 weeks with home blood pressure readings. Medication dose adjustment would be considered based on response and tolerance. Repeat fasting glucose, HbA1c, lipid profile, kidney function, and urine albumin-creatinine ratio were advised in 3 to 6 months depending on clinical progress.

The patient was advised to complete fundus examination and continue routine preventive health screening.

Patient Education Summary

Hypertension may not cause symptoms, but it can damage blood vessels over time. Regular monitoring and consistent treatment reduce the risk of stroke, heart disease, kidney disease, and eye complications. Lifestyle changes are part of treatment and should continue even when medication is started.

Medication should not be stopped suddenly without consulting a clinician. Side effects such as ankle swelling, dizziness, flushing, or unusual fatigue should be reported during follow-up.

Discharge And Safety Instructions

The patient was stable at the end of the visit and was discharged from observation with advice. She was instructed to return earlier if symptoms worsened or if red-flag symptoms appeared. She was also advised not to self-medicate for persistent headache and not to ignore repeated high blood pressure readings.

Doctor's Note

This case report is intended for MediVault AI testing. It contains enough structured information to test retrieval of symptoms, investigations, diagnosis, treatment, lifestyle advice, red flags, and follow-up plans. It is fictional and should not be used as medical advice.